Software Requirements Specification

for

College Internship Tracking and Management System

Version 1.0

Prepared by

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1 Introduction

The **College Internship Tracking and Management System (ITMS)** is a software solution designed to help colleges manage internship opportunities and student participation efficiently. Many colleges currently rely on spreadsheets, emails, and manual tracking methods to manage internship activities. These methods often lead to data inconsistencies, poor coordination, and difficulty in monitoring student internship status.
The proposed system aims to develop a **centralized database application** that stores internship opportunities, student profiles, company details, and application records. The system allows students to apply for internships, companies to post internship opportunities, and faculty coordinators to monitor internship participation.

1.1 Document Purpose

This document specifies the **software requirements** for the Internship Tracking and Management System. It defines both the functional and non-functional requirements of the system and provides a clear understanding of the system's features and constraints.
The SRS acts as a reference document for developers and evaluators involved in the development and implementation of the system.

1.2 Product Scope

The **Internship Tracking and Management System** is designed to simplify internship management by providing a centralized platform where students, companies, and faculty members can interact.
The system will:

- Maintain student internship profiles
- Allow companies to post internship opportunities
- Enable students to apply for internships
- Track application status
- Allow faculty coordinators to monitor internship activities

By digitizing internship management, the system improves transparency, coordination, and data accuracy.

1.3 Intended Audience and Document Overview

This document is intended for developers, testers and faculty coordinators involved in the development and use of the Internship Tracking and Management System. It also serves as a reference for project evaluators and instructors to assess the system’s scope, requirements, and design decisions. Each reader can use this document to gain a clear understanding of the system’s objectives, functionalities, and operational constraints.
The Software Requirements Specification (SRS) is organized to provide a structured and detailed description of the system. It begins with an **Introduction**, which presents the purpose, scope, and intended audience of the document. The **Overall Description** section explains the system’s main

features, user interactions, and assumptions related to its operation. The **Specific Requirements** section defines the functional and interface requirements needed to implement the system effectively, including user interactions and system behavior. The **Non-functional Requirements** section outlines important system qualities such as performance, security, and reliability. Finally, the appendices include supporting information such as the data dictionary and group activity records.

For effective use of this document, readers should begin with the **Introduction** to understand the overall objectives of the system. Developers and testers should focus primarily on the **Specific Requirements** and **Non-functional Requirements** sections to guide system implementation and testing. Faculty coordinators may find the **Overall Description** and appendices more useful for understanding the system's capabilities, structure, and operational constraints.

1.4 Definitions, Acronyms and Abbreviations

● **SRS**: Software Requirements Specification
● **ITMS**: Internship Tracking and Management System
● **DBMS**: Database Management System
● **ER Model**: Entity Relationship Model used for database design
● **UML**: Unified Modeling Language
● **GUI**: Graphical User Interface
● **API**: Application Programming Interface
● **SQL**: Structured Query Language used to manage relational databases
● **CSV**: Comma Separated Values file format used for data export
● **Admin**: Authorized user responsible for managing the system and maintaining records
● **Student**: A registered user who can view and apply for internship opportunities
● **Company**: An organization that offers internship opportunities to students
● **Internship**: A temporary work opportunity provided by a company for students
● **Application**: A record representing a student's request to participate in a specific internship
● **Application Status**: The current state of an internship application (e.g., Pending, Accepted, Rejected)
● **Faculty Coordinator:** A faculty member responsible for monitoring internship activities and student participation
● **Internship Listing**: A posted internship opportunity containing details such as role, duration, and eligibility

1.5 Document Conventions

This SRS follows IEEE documentation guidelines. The text is written in Arial font, size 11, with single spacing and 1-inch margins. Section and subsection headings use a bold, numbered format for clarity and consistency. Italic text may be used for comments or examples. Variable names and database attributes follow camelCase notation (e.g., studentID, internshipID). Functional requirements are labeled with the prefix "F" followed by a number (e.g., F1, F2) for easy reference.

2 Overall Description

2.1 Product Overview

The Internship Tracking and Management System (ITMS) is a centralized application designed to streamline the management of internship opportunities, student applications, and company participation within a college. The system provides a unified platform where students can view available internships, apply for opportunities, and track their application status. Faculty coordinators and administrators can monitor internship participation and manage internship records efficiently. By replacing manual tracking methods such as spreadsheets and emails, the system offers a more organized, efficient, and reliable approach to managing internship data.

The system functions as a centralized database application that manages interactions between students, companies, and faculty coordinators. It supports key modules such as student management, internship listings, application tracking, and reporting. The platform can also support future integration with external services such as email notifications or analytics tools. This modular design ensures that the system can scale easily as the number of users and internship opportunities increases.

2.2 Product Functionality

The Internship Tracking and Management System provides essential functionalities to streamline the process of managing internship opportunities and student participation within a college. The major functions of the system include:
● **Student Management:** Allows students to create and update their profiles, including academic details and contact information.
● **Internship Listings Management:** Enables companies or administrators to post, update, and remove internship opportunities with details such as role, duration, and eligibility criteria.
● **Application Management:** Allows students to apply for internships and enables the system to store and manage application records.
● **Application Status Tracking:** Tracks the status of each application (e.g., pending, accepted, rejected) and allows students to monitor their progress.
● **Faculty Monitoring:** Allows faculty coordinators to view internship applications, monitor student participation, and maintain internship records.
● **Reporting and Analytics:** Generates reports related to internships, such as the number of applications, selected students, and internship participation statistics.

2.3 Design and Implementation Constraints

The **Internship Tracking and Management System** is subject to several constraints that influence its design and development. The system must follow **standard database design principles** and **structured software development practices** to ensure maintainability and reliability. Additionally, **UML (Unified Modeling Language)** may be used for system modeling, allowing clear visualization of **system components, user interactions, and data relationships**. The system will utilize **MySQL** as the database for storing and managing data related to **students, companies, internships, and applications**. The application may be implemented as a **web-based system** accessible through standard web browsers. The system must implement

security mechanisms such as user authentication and controlled access to ensure that only authorized users can modify system data. The system should also be designed with **scalability** in mind so that future enhancements such as **notification systems, analytics, or additional reporting features** can be incorporated easily.

References:

● **Elmasri, R., & Navathe, S. (2016).** *Fundamentals of Database Systems*. Pearson Education.
● **Booch, G., Rumbaugh, J., & Jacobson, I. (2005).** *The Unified Modeling Language User Guide*. Pearson Education.

2.4 Assumptions and Dependencies

● Assumes **students are registered with valid college identification details** before accessing the system.
● Assumes **companies or administrators provide accurate internship information**, including role descriptions and requirements.
● Depends on a **relational database management system (e.g., MySQL)** for storing and managing system data.
● Assumes **stable internet connectivity** for accessing the system and submitting internship applications.
● Assumes **faculty coordinators are available to monitor internship applications and student participation** within the system.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The ITMS features a desktop interface with a dashboard and pages for inventory, sales, and billing. Key elements include:

1. Login Page:
    a. A secure login page where users such as students, faculty coordinators, or administrators enter their **username and password** to access the system.
2. Dashboard:
    a. Provides navigation to different sections such as **student profiles, internship listings, applications, and reports**.
3. Student Profile Page:
    a. Allows students to **view and update their personal and academic details**.
    b. Students can manage their **profile information and internship preferences**.
4. Internship Listings Page:
    a. Displays available **internship opportunities posted by companies or administrators**.
    b. Students can **view internship details such as role, duration, eligibility, and company information**.
    c. Students can **submit applications directly from this page**.
5. Application Tracking Page:
    a. Allows students to **view the status of their submitted applications.**

6. Admin / Faculty Dashboard:
    a. Allows administrators or faculty coordinators to **manage internship postings and monitor student applications**.
    b. Provides access to **system reports and internship participation records**.

3.1.2 Hardware Interfaces

1. Personal Computer / Laptop
    - Used by students, faculty coordinators, and administrators to access the system through a web browser.
    - Handles user interaction and system operations.
2. Internet Connectivity
    - Required to access the system and communicate with the database server.
3. Server System
    - Hosts the application and database for storing internship-related information.

3.1.3 Software Interfaces

1. Database System
   - The system interacts with a relational database (e.g., **MySQL**) to store and retrieve data related to **students, companies, internships, and applications.**
   - SQL queries are used for data manipulation and retrieval.
2. Operating System
   - The system can run on **Windows, Linux, or macOS** environments and interacts with system-level services for application execution.
3. Web Browser
   - Users access the system through standard web browsers such as **Google Chrome, Mozilla Firefox, or Microsoft Edge**.
4. Email Notification Services
   - The system may integrate with email services to send **notifications about internship updates or application status changes**.
5. Backup & Export Tools
   - The system may provide options to **export reports in CSV or PDF format** for record keeping and analysis.
   - Automated or manual backups can be implemented to prevent data loss.

3.2 Functional Requirements

3.2.1 F1 : The system shall maintain student records

The system shall store and manage detailed **student information** in a relational database, including student ID (unique identifier), name, department, academic year, email address, and contact number. Students shall be able to view and update certain profile details through the system interface.

3.2.2 F2 : The system shall maintain company records

The system shall store and manage **company information** in the database, including company name, contact email, location, and internship details. Administrators shall have the ability to add, update, or remove company records when necessary.

3.2.3 F3 : The system shall allow companies or administrators to post internship opportunities

The system shall provide functionality for authorized users to:

- **Add:** Create new internship listings with details such as role, description, duration, eligibility requirements, and company information.
- **Update:** Modify internship details if changes occur.
- **Delete:** Remove internship listings that are no longer available.

3.2.4 F4 : The system shall allow students to apply for internships

Students shall be able to submit internship applications through the system. Each application shall contain information such as student ID, internship ID, application date, and supporting documents if required. The system shall store these applications in the database for tracking and review.

3.2.5 F5 : The system shall track application status

The system shall maintain the status of each internship application. Possible statuses include:

- Pending
- Accepted
- Rejected

Students shall be able to view the status of their applications through the system interface.

3.2.6 F6 : The system shall generate internship reports and analytics

The system shall generate reports related to internship activities, including:

- Number of internships available and applied for
- Number of selected students
- Internship participation statistics by department or company

Reports may be exported in formats such as **PDF or CSV** and viewed through the administrative dashboard.

4 Other Non-functional Requirements

4.1 Performance Requirements

Performance requirements ensure that the **Internship Tracking and Management System (ITMS)** operates efficiently and provides a smooth user experience when accessing internship information and submitting applications. These requirements support the functional requirements described in **Section 3.2** and help maintain acceptable system responsiveness.

- **P1**: The system shall allow users to view available internship listings quickly when requested.
  Rationale: Fast access to internship opportunities allows students to easily browse available internships without unnecessary delays.
- **P2**: The system shall store student internship applications promptly after submission.
  Rationale: Efficient processing of applications ensures that student submissions are recorded correctly and prevents loss or duplication of application data.
- **P3**: The system shall update application status changes promptly when modified by administrators.
  Rationale: Timely updates ensure that students and faculty coordinators can view the latest application status without confusion.
- **P4**: The system shall generate internship reports within a reasonable time even when the system contains many records.
  Rationale: Efficient report generation helps administrators analyze internship participation and system usage effectively.

4.2 Safety and Security Requirements

These requirements address potential risks such as **data loss, unauthorized access, and misuse of internship information**, ensuring that the system protects sensitive data related to **students, companies, and internship applications**. Safeguards are defined to maintain data integrity and ensure that only authorized users can access or modify system records.

- **S1**: The system shall require user authentication via username and password for all users accessing the system.
  Rationale: Login authentication prevents unauthorized users from accessing or modifying student records, internship postings, and application information.
- **S2**: The system shall automatically back up the database regularly to a secure storage location.
  Rationale: Regular backups protect against data loss caused by hardware failure, accidental deletion, or system errors. Backup files should be stored securely and retained for a defined period to allow recovery of system data when needed.
- **S3**: The system shall record important user actions with timestamps and user identification.
  Rationale: Keeping logs of actions such as posting internships, submitting applications, or updating application status helps administrators monitor system activity and detect possible errors or misuse.
- **S4**: The system shall implement error handling mechanisms in database operations to prevent data corruption.

Rationale: Proper error handling ensures that the system handles problems safely and prevents incomplete or corrupted data from being stored in the database.

4.3 Software Quality Attributes

Software quality attributes describe important system characteristics beyond functionality and performance. These attributes focus on ensuring that the **Internship Tracking and Management System (ITMS)** is reliable and flexible for students, faculty coordinators and administrators.

4.3.1 Reliability

Requirement: The system should remain available and function properly during normal usage, with minimal downtime.

How Achieved:

- Implement reliable **database connections and backup mechanisms** to prevent data loss in case of system failures.
- Implement proper error handling to ensure the system continues to operate even if unexpected errors occur.
- Perform **system testing with multiple concurrent users** to ensure stable performance when many students access internship listings or submit applications simultaneously.

Rationale: Reliable system performance ensures that students and administrators can access internship information and submit applications without interruptions.

4.3.2 Adaptability

Requirement: The system should allow new features or improvements to be added without major changes to the overall system design.

How Achieved:

- Use a modular database structure with separate tables for students, companies, internships, and applications.
- Design the system so that new pages or features can be added easily in the future.
- Maintain clear documentation of the system structure and database design.

Rationale: Colleges may introduce new internship requirements or additional features over time. A flexible system design allows the platform to adapt to these changes while maintaining stable operation.

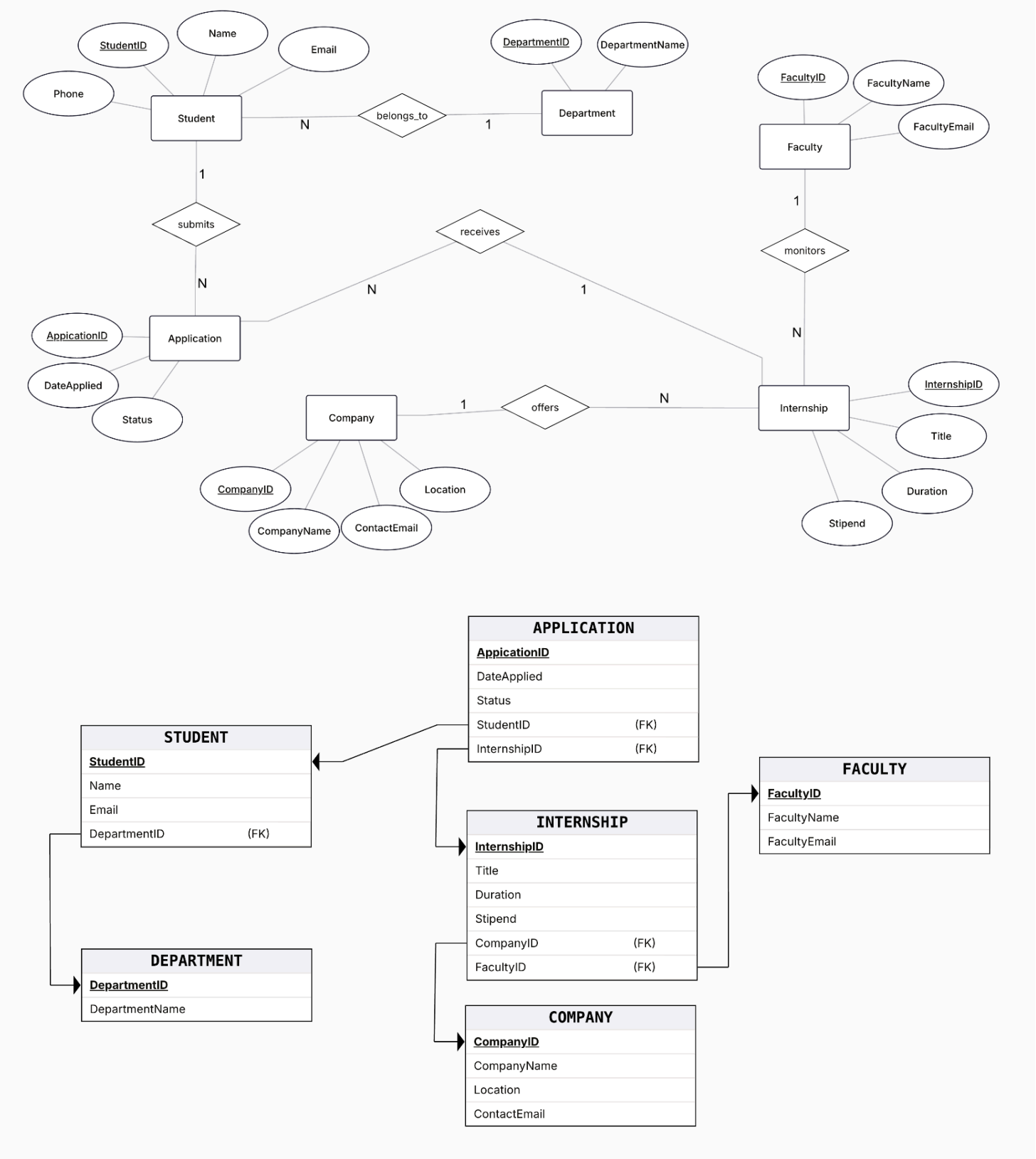
Appendix A – Data Dictionary

This section lists the key variables, attributes, and data elements used in the Internship Tracking and Management System. It describes their meanings, possible values, and the operations in which they are used.

Variable/State	Description	Possible Values	Related Operations
StudentID(PK)	Unique identifier for each student(referenced in Application to indicate the applicant)	Integer (Auto-increment)	Register Student, View Profile
Name	Name of the student	String (Max 50 chars)	Register Student, Update Profile
Email	Email address of the student	String (Email format)	Register Student, Update Profile
Phone	Phone number of the student	String (10–15 digits)	Register Student, Update Profile
DepartmentID(PK)	Unique identifier for each department(reference d in Student to indicate the student's department)	Integer (Auto-increment)	Add Department
DepartmentName	Name of the department	String (Max 50 chars)	Add Department, Update Department
FacultyID(PK)	Unique identifier for faculty(referenced in Internship to indicate the supervising faculty)	Integer (Auto-increment)	Add Faculty
FacultyName	Name of the faculty supervisor	String (Max 50 chars)	Add Faculty, Update Faculty
FacultyEmail	Email address of faculty	String (Email format)	Add Faculty
CompanyID(PK)	Unique identifier for each company(referenced by Internship to indicate the company	Integer (Auto-increment)	Add Company, Update Company

	offering the internship)		
CompanyName	Name of the company offering internships	String (Max 50 chars)	Add Company, Update Company
ContactEmail	Official contact email of company	String (Email format)	Add Company
Location	Location of the company	String (Max 100 chars)	Add Company
InternshipID(PK)	Unique identifier for each internship opportunity(referenced in Application for internship applications)	Integer (Auto-increment)	Post Internship, Update Internship
Title	Title of the internship	String (Max 100 chars)	Post Internship
Stipend	Payment offered for internship	Integer / Decimal	Post Internship
Duration	Duration of the internship	String (e.g., 2 months, 6 months)	Post Internship
ApplicationID(PK)	Unique identifier for each internship application	Integer (Auto-increment)	Submit Application, Track Application
DateApplied	Date when the internship application was submitted	Date	Submit Application
Status	Current status of the application	Pending, Accepted, Rejected	Update Application Status, View Application

Appendix B – ER Model



Appendix C – Group Log

Date	Participants	Activities
08-03-2026	All members	Initial discussion of project idea and system scope
09-03-2026	Harikeerthan Devan, Shuaib F	Requirement analysis and system functionality discussion
10-03-2026	Govind Menon K, Muhammed Fahad K	Prepared initial SRS draft and document structure
11-03-2026	Aman Ashraf, Swathy V V	Designed ER diagram and reviewed entity relationships
12-03-2026	Harikeerthan Devan, Shuaib F	Prepared database schema and data dictionary
13-03-2026	Govind Menon K, Muhammed Fahad K	Reviewed functional and non-functional requirements
14-03-2026	All members	Final review, formatting, and submission